

Testing Types and Techniques

From Smoke and Regression to UAT, Exploratory, performance and test design techniques.

Test levels

Testing can be organized into levels such as component/unit, integration, system and acceptance. Each level targets different classes of problems and should be understood in the context of product architecture and lifecycle.

Smoke Testing

Smoke testing is a short check of the most critical paths to decide whether a build is stable enough for further testing. In e-commerce this may include Home, search, PDP, cart and the beginning of checkout.

Regression Testing

Regression checks whether changes caused unintended effects in existing functionality. Selection should consider risk, criticality, changed areas, integrations, defect history and available time.

Retest vs Regression

Retest specifically confirms whether a fixed defect no longer occurs. Regression checks whether the fix or another change affected other areas. Both are often necessary after a fix.

UAT - User Acceptance Testing

UAT validates whether the solution meets business needs and acceptance expectations from the perspective of business users, customer representatives or final/representative users. QA may support environment, data and evidence, but UAT should not simply be treated as another QA test cycle.

Exploratory Testing

In exploratory testing, learning, design and execution are integrated. Charters, notes, heuristics and debriefing help make exploration intentional, traceable and risk focused.

Performance: load, stress, spike and soak

Load evaluates behavior under planned load; Stress pushes beyond normal limits; Spike observes sudden changes; Soak/Endurance evaluates prolonged behavior. Performance is a broad category and these tests answer different questions.

Test design techniques

Equivalence partitioning, boundary value analysis, decision tables and state transition testing help select tests systematically. Pairwise and heuristics such as Goldilocks can complement analysis depending on the problem.

Goldilocks

The Goldilocks heuristic encourages testing something that is too small, just right and too big. For a 10 MB upload limit, explore an empty file, a valid mid-sized file and a file over the limit. It is a thinking aid, not a replacement for formal boundary analysis.

How to choose a test type?

Start with the question and risk. If you need to know whether a build is testable, think Smoke. If a change may affect existing behavior, Regression. If you need behavior under many users, Performance. The test label should serve the objective, not the other way around.

QA tip

100% of test cases executed does not mean 100% of the product tested: think in terms of risk and coverage.